

Potential Improvements for Cross-Season Sentinel-2 Image Matching

LoFTR is an excellent detector-free matcher that handles low-texture areas well. LoFTR is computationally heavy due to its dense attention mechanisms. For faster inference over massive datasets, evaluating sparse feature-matching architectures such as SuperPoint coupled with LightGlue or SuperGlue will probably work better. These networks are often more lightweight and can accelerate inference times without sacrificing the geometric accuracy required for satellite imagery.